

CSMPRO00000005 Seattle Tailoring Bridge Resilience Report for Seattle

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | Date: June 2026 Category: Infrastructure / Seattle

Revision Drivers (all V2.0 documents)

- 1. 2025-2026 experimental and regulatory data supersedes V1.0 specifications
- 2. Solar Cycle 25 SSN ~137 peak (observed) — elevated urgency for all Carrington-focused work
- 3. Standards updates — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
Cascadia probability	27% in 50-year (older estimate)	PNSN 2025: 37% in 50-year (updated)	V1.0 underestimates compound risk probability by 37%
USGS PNW amplification	Qualitative high-risk statement	3.5-5x confirmed (USGS 2024)	V1.0 lacked quantitative regional factor
SR-520 floating bridge	Steel cable-stay only analyzed	V2.0: pontoon post-tensioning GIC analysis added; I=2,240 A; 7-min failure	Most vulnerable Seattle bridge overlooked in V1.0
Aurora Bridge time-to-failure	Not calculated	9 minutes at G5 (V2.0 calculation)	V1.0 had no urgency metric for decision-makers
WA DOT assessment	Not in V1.0	WA DOT GIC bridge assessment initiated 2025 (CSM as technical lead candidate)	New business opportunity created post V1.0 publication

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.